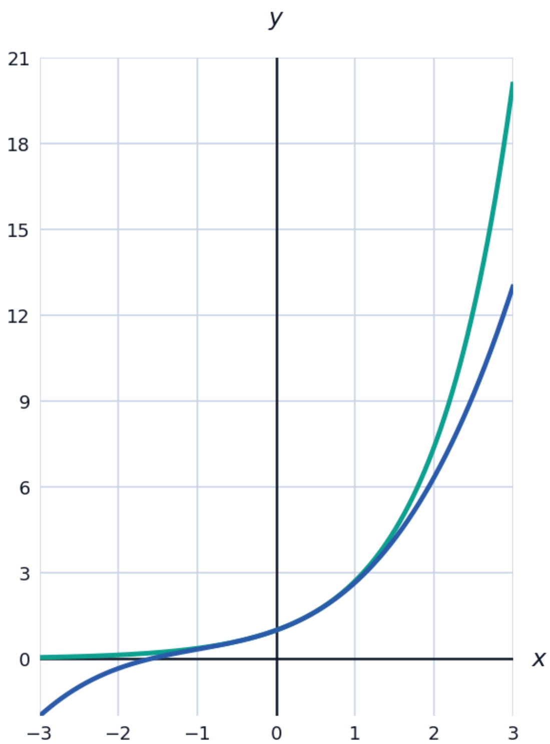
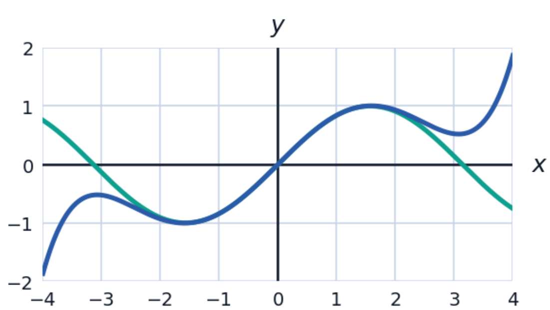
Maclaurin Series & Convergence of Series

Constructing a Maclaurin series

1 Find the first four non-zero terms of the Maclaurin series for $f(x) = e^x$, using $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$.



2 Find the Maclaurin series for $f(x) = \sin x$ up to the term in x^5.



Using known series to build new ones

3 Using the Maclaurin series for e^x, find the Maclaurin series for e^{-2x} up to the term in x^3.

4 Given the Maclaurin series $\sin x \approx x - \frac{x^3}{6}$ and $\cos x \approx 1 - \frac{x^2}{2}$, find the Maclaurin series for $\tan x$ up to the term in x^3, using $\tan x = \frac{\sin x}{\cos x}$ and polynomial division.

Using a Maclaurin series to approximate a value

5 Use the Maclaurin series for e^x up to the x^3 term to approximate $e^{0.1}$, to four decimal places, and compare with the true value $e^{0.1} \approx 1.10517$.

Tests for convergence of a series

6 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

7 Use the ratio test to determine whether the Maclaurin series for e^x, $\sum_{n=0}^{\infty} \frac{x^n}{n!}$, converges for all real x.

Determining the radius of convergence

8 Use the ratio test to find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{x^n}{n}$.

9 Find the radius of convergence of $\sum_{n=0}^{\infty} n!\, x^n$, and explain what this tells us about where the series is useful.

Comparing a Maclaurin approximation to the exact function

10 Using the Maclaurin series $\cos x \approx 1 - \frac{x^2}{2} + \frac{x^4}{24}$, approximate $\cos(0.3)$ to four decimal places, and compare with the true value $\cos(0.3) \approx 0.9553$.

Maclaurin series for ln(1+x)

11 Find the Maclaurin series for $f(x) = \ln(1 + x)$ up to the term in x^4, using $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$.

Determining the interval of convergence using the ratio test

12 Find the radius of convergence of the Maclaurin series for $\ln(1 + x)$, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}x^n}{n}$, using the ratio test.

Combining Maclaurin series through multiplication

13 Using the Maclaurin series $e^x \approx 1 + x + \frac{x^2}{2}$ and $\frac{1}{1-x} \approx 1 + x + x^2$ (valid for small x), find the Maclaurin series for $\frac{e^x}{1-x}$ up to the term in x^2, by multiplying the two series together.

Using differentiation to find a Maclaurin series

14 Given the Maclaurin series $\frac{1}{1-x} \approx 1 + x + x^2 + x^3 + \cdots$ for $|x| < 1$, differentiate both sides term by term to find the Maclaurin series for $\frac{1}{(1-x)^2}$ up to the term in x^2.

Estimating an error bound in a Maclaurin approximation

15 The Maclaurin series for $\sin x$ truncated after the x^3 term is $\sin x \approx x - \frac{x^3}{6}$. Use this to approximate $\sin(0.2)$ to four decimal places, and find the absolute error compared to the true value $\sin(0.2) \approx 0.19867$.

16 Find the first three non-zero terms of the Maclaurin series for $f(x) = (1+x)^{1/2}$, using the binomial series $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$ with $n = \frac{1}{2}$.

17 Use the Maclaurin series for e^x up to the x^4 term to approximate $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$, and check that your answer matches the value obtained using L'Hopital's rule.

18 This question has three parts. (a) Find the Maclaurin series for $f(x) = \cos(x^2)$ up to the term in x^8, by substituting x^2 into the known series $\cos u \approx 1 - \frac{u^2}{2} + \frac{u^4}{24}$. (b) Use your result to write down an approximation for $\int_0^1 \cos(x^2)\,dx$ as a sum of simple fractions (this integral has no elementary closed form, so series approximation is genuinely useful here). (c) Evaluate this approximation as a single decimal, to three decimal places.